

HDI Predictor Documentation

Initial Project Planning

Date	17 June 2025
Team ID	SB-GENAI-HDI-2025
Project Name	HDI Predictor - Human Development Index Classification System
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Product Backlog, Sprint Schedule, and Estimation

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Data Preparation	USN-1	As a developer, I can collect and clean a country-level HDI indicator dataset (life expectancy, schooling, GNI per capita)	2	High	17 Jun 2025	20 Jun 2025
Sprint-1	Data Preparation	USN-2	As a developer, I can perform EDA to understand indicator distributions and correlations with HDI tier	2	High	17 Jun 2025	20 Jun 2025
Sprint-1	Model Building	USN-3	As a developer, I can train and compare classification models (Logistic Regression, RandomForest) on the indicator dataset	3	High	20 Jun 2025	23 Jun 2025
Sprint-2	Model Building	USN-4	As a developer, I can tune the RandomForestClassifier and validate it reaches $\geq 95\%$ test accuracy	3	High	23 Jun 2025	25 Jun 2025
Sprint-2	Model Building	USN-5	As a developer, I can serialize the trained model as <code>model.pkl</code> for reuse at inference time	1	Medium	23 Jun 2025	25 Jun 2025
Sprint-2	Web Application	USN-6	As a user, I can enter the four HDI indicators into a Flask web form	2	High	25 Jun 2025	28 Jun 2025
Sprint-2	Web Application	USN-7	As a user, I can submit the form and see the predicted HDI tier and score rendered on the page	2	High	25 Jun 2025	28 Jun 2025

Sprint-3	Explainability	USN-8	As a user, I can see which feature most influenced my prediction	2	Medium	28 Jun 2025	30 Jun 2025
Sprint-3	Deployment	USN-9	As a developer, I can containerize the app with Docker so it runs identically for any evaluator	2	High	30 Jun 2025	02 Jul 2025
Sprint-3	Testing & Documentation	USN-10	As a developer, I can test the deployed app end-to-end and document setup/run instructions	2	High	02 Jul 2025	04 Jul 2025

Sprint Summary

Sprint	Total Story Points	Focus
Sprint-1	7	Dataset collection, cleaning, and exploratory analysis
Sprint-2	8	Model training, serialization, and core Flask integration
Sprint-3	6	Explainability, Docker deployment, and final testing/documentation